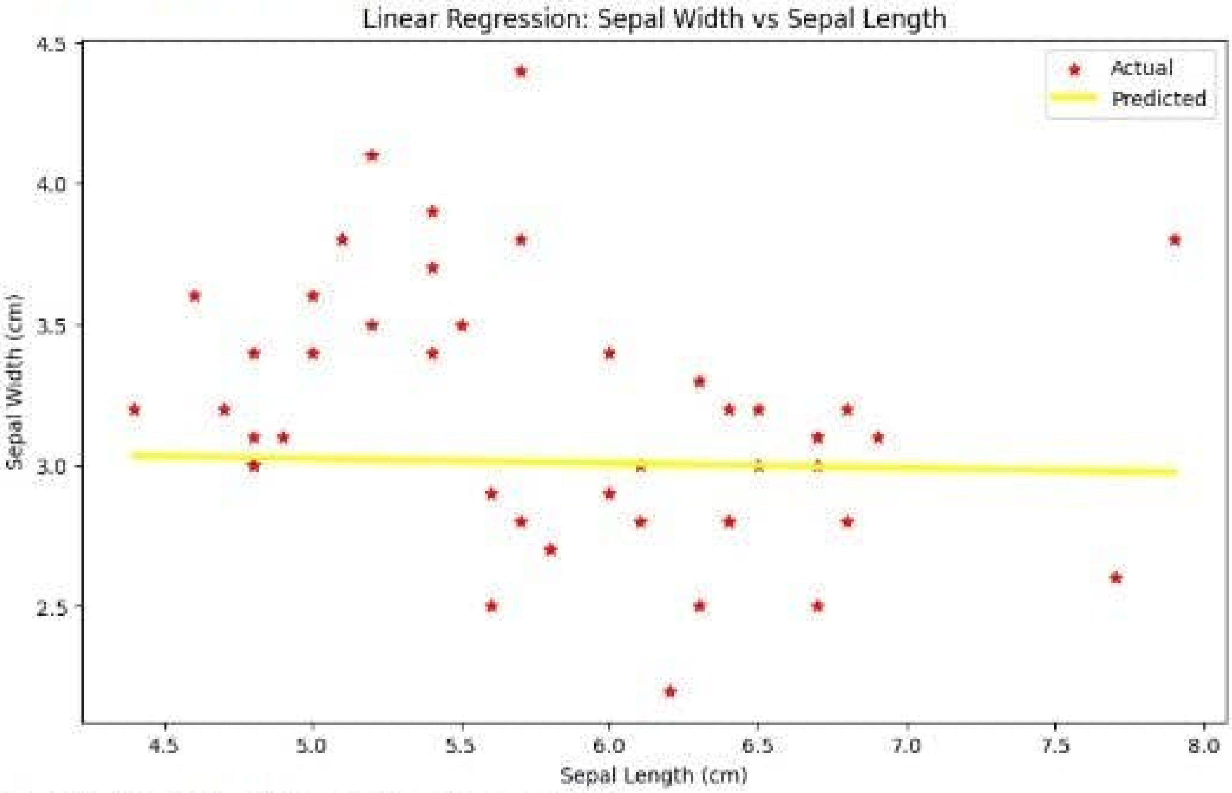
```python
X_test_sorted = X_test.iloc[sorted_index]
y_pred_sorted = y_pred[sorted_index]

# Plot
plt.figure(figsize=(10, 6))
plt.scatter(X_test, y_test, color='red', marker='*', label='Actual')
plt.plot(X_test_sorted, y_pred_sorted, color='yellow', linewidth=3, label='Predicted')

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Linear Regression: Sepal Width vs Sepal Length')
plt.legend()

plt.show()

# Predict for a new sample
new_sample = pd.DataFrame([[5]], columns=['sepal length (cm)'])
predicted_width = model.predict(new_sample)

print(
    f"The predicted sepal width for sepal length "
    f"{new_sample.iloc[0,0]} cm is {predicted_width[0]:.2f} cm"
)
```

```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   species
0        0
1        0
2        0
3        0
4        0
Mean Squared Error on test set: 0.23
```




```
The predicted sepal width for sepal length 5 cm is 3.02 cm
```